Deferred Forests and Exceptional Incidence

Exact local reductions for compressed unknot hierarchies

Research continuation for Vladimir Reshetnikov's ProveIt project

Prepared with ChatGPT Pro; independent mathematical review invited

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Repository baseline: `c6d0ec8aa0b6308310ab6c081ba29d8bc363ede5`

Abstract

We eliminate one exponential search variable in the repository's raw geometric handle-cancellation problem. If two-handle rows are literal closed walks in the zero/one-handle graph, column-leaf peeling on the full incidence matrix preserves the original graph components when its graph pivots are deleted. A spanning forest can therefore be chosen *after discovery*, avoiding every pivot, and contracted *before replay*. This is an exact test over all primal forests for complete raw positive-singleton cancellation; a stopping set provides a universal refusal witness for this restricted model. The result also applies to the authenticated finite bundle-boundary matrix when all annular rows are assembled first.

For the remaining dual-forest choice, we prove an exact finite selector with running time $2^{O(u\log(u+2))}\,\mathrm{poly}(N)$, where u counts rows incident with active columns supported on more than two rows. The proof compresses ordinary implications onto this exceptional interface and enumerates partial set partitions, using at most $\mathrm{Bell}(u+1)$ polynomial graphic/partition matroid-intersection calls. For the original native core columns, the established three-occurrence bound implies $u \le 3k$.

An additive implementation increases historical whole-cut certificate coverage from 92/106 to 99/106, and additional-case coverage from 10/14 to 12/14, with all 213 old/new positive outputs independently replayed. The standalone dual selector agrees with exhaustive oracles on 50,658 finite instances. We supply full proofs, a geometric all-size separation family, raw experiments, source-bound certificates, a patch, and a prioritized research programme. These are local mathematical and implementation advances. We do not claim a complete quasi-polynomial unknot recognizer or a global logarithmic exceptional-interface bound.

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1 Scope, provenance, and the precise contribution

The engineering objective is to make compressed normal cuts useful inside an eventual fast hierarchy algorithm for unknot recognition. A terminal operation already present in ProveIt takes a triangulation and a normal surface in binary coordinates, builds source-authenticated compressed geometry, and tries to certify the genera and multiplicities of all handlebody components of the cut. Its output is deliberately partial: a successful certificate is independently replayable; a failed search is `INCONCLUSIVE`. The operation is not itself a decision procedure for all knots, nor does a handlebody census alone encode every marked boundary map needed to continue a hierarchy.

We work at commit

`c6d0ec8aa0b6308310ab6c081ba29d8bc363ede5`.

The inspected material includes the maintained `fast` implementation, the synthesis notes, and all five archives in `docs/incoming` at this commit. The delivered patch is against that revision, not a moving branch. The source audit records the hashes of the incoming archives and changed modules. No repository publication or merge is part of this package.

1.1 What the preceding research already established

The incoming material is substantial. In particular, the following are inputs to this continuation, not new claims made here.

1. *Positive cancellation.* Complete elimination of the rows of a nonnegative occurrence matrix by positive singleton pivots is equivalent to column-leaf peeling. A greatest stopping set is the obstruction, and immutable source entries admit a sparse queue and timestamp replay [12]. We reprove the required statement in Section 2 so the new argument is self-contained.
2. *Dual normalization.* The native top handles can be removed using a source-authenticated dual spanning tree, relative to the exposed ports. For active columns on at most two rows, the remaining tree-selection problem has a polynomial graphic/partition matroid formulation. Original native source columns have at most three literal occurrences; their odd supports obey the dual cycle condition [9].
3. *Other compressed infrastructure.* Component gauges, cut atlases, occurrence-preserving flag regularization, weighted forest heuristics, and homological progress certificates were developed in the other incoming notes [8, 11, 10]. They remain relevant to successor construction and to other move classes.

The maintained code at our baseline already incorporates several of these ideas, including a deterministic breadth-first dual-tree reduction. Its historical whole-cut census coverage is 92/106. Reimplementing that reduction or rediscovering the matrix-only peeling theorem would not be a new contribution.

1.2 New results relative to the inspected baseline

Our first result uses geometry that the matrix-only model discarded: attaching curves are closed walks in the primal graph. It proves that all useful peeling pivots can be kept outside some spanning forest. Thus the primal forest can be eliminated as a search variable, even though its canonical choice can destroy all apparent progress. The constructive proof gives a

strictly larger native certificate class without introducing word reduction, speculative handle slides, or a new topology oracle.

Our second result addresses the remaining high-support columns in dual selection. Their incident rows form a finite interface. Ordinary implications can be compressed onto that interface, and a successful tree needs at most one additional witness for each interface row. Enumerating *how targets share witnesses*, rather than enumerating all choices of tree edges, gives a Bell-number parameter dependence. The classical matroid machinery is reused; the exceptional-interface reduction is the new part.

A third contribution is the exact geometric separation family of Section 4. It identifies a genuine failure of a fixed forest policy and an exponential candidate space, while making no lower-bound claim against every possible algorithm. The implementation and the native experiments then test whether the first theorem helps the actual source geometry. It does: nine additional whole-cut packets are certified across the two supplied corpora.

Here and throughout, "new" means absent from the inspected repository and incoming research and developed in this continuation. We have not established a comprehensive priority claim across all of low-dimensional topology, hypergraph peeling, matroid activation, and sparse elimination. The mathematical statements are proved in the article and subjected to separate proof review and implementation audits; they have not been formalized in a proof assistant or externally peer reviewed.

1.3 Relation to Lackenby's announced target

The 2021 talk materials announce $n^{O(\log n)}$ unknot recognition; an Oxford compressed handout separately displays a weaker $n^{O(\log^2 n)}$ estimate. Both are quasi-polynomial, but their exponents should not be conflated [4, 5]. Lackenby's July 2026 paper gives the hierarchy algorithm and explicitly leaves speed-up to future work [6]. Its Theorem 6.2 already constructs a nonseparating oriented normal surface in polynomial time under its handle hypotheses. Proposition 9.1 bounds the relevant loop count by $L(g+1)^L$; the refined construction has length at most $4c_q$. Theorem 11.3 controls uniform handle type. Proposition 14.2 and Theorem 14.4 give compressed successor and parallelity-bundle constructions under their stated geometric hypotheses. These are established local results, not open problems solved by this package.

Our target is to turn applicable theory into a source-authenticated, bounded-cost implementation and to improve the local terminal and normalization operators. Neither a faster terminal census nor a parameterized forest selector supplies the missing global hierarchy bound. Section 10 writes those remaining obligations explicitly.

1.4 Parameters and trust boundaries

We distinguish the crossing count n of a knot diagram from the encoded size N of a particular finite graph/matrix or compressed cut state. A normal weight W is different again: its bit length is logarithmic in W. A polynomial in expanded normal weight is generally unusable for the desired asymptotic goal.

The finite theorems accept exact integer matrices and graphs. A topology application must additionally authenticate that these objects arise from the specified handles, attaching charts, and legal port-preserving moves. A matrix with the right parity is not automatically a native normal cut. This distinction applies especially to the triangle witness in Section 5: it tests a constrained finite class, while the nine new native certificates have full source records and independent whole-cut replay.

2 The raw cancellation invariant

This section records the exact algebraic prerequisite from the preceding positive-cancellation work [12]. The use of unsigned literal occurrences is essential. These counts record intersections with specified belt discs; they are not exponent sums of freely reduced words.

Definition 2.1 (Positive singleton pivot and stopping set)**.** Let $A = (a_{rc})$ be a finite nonnegative integer matrix with rows R and columns C. A positive singleton pivot chooses (r, c) with $a_{rc} = 1$, deletes row r and column c, and changes each surviving entry to

$$a'_{id} = a_{id} + a_{ic}a_{rd} \qquad (i \neq r,\ d \neq c).$$

A nonempty set $S \subseteq R$ is a *stopping set* if

$$s_c := \sum_{r \in S} a_{rc} \in \{0\} \cup \{2, 3, \ldots\} \quad \text{for every } c \in C.$$

A *column leaf* has total live column occurrence one. Pivoting on it creates no fill in the surviving rows and is called a peeling move.

The formula deliberately has a plus sign. It is the conservative raw intersection update: algebraic cancellation of opposite letters is not part of this model. The theorem below cannot be extended to arbitrary subtractive elimination merely by reusing its proof.

Lemma 2.2 (A stopping set survives positive pivots)**.** *If A contains a stopping set, then every legal positive singleton pivot leaves a nonempty stopping set on surviving rows.*

Proof. Let (r, c) be the pivot and let S be a stopping set. If $r \notin S$, retain S. For each surviving column d its new total is $s_d + s_c a_{rd}$. If this is positive, either $s_d \geq 2$ or $s_c a_{rd} \geq 2$, so it is not one.

If $r \in S$, the pivot entry is one and $s_c \neq 1$, hence $s_c \geq 2$. Another row of S meets c, so $S' = S \setminus \{r\}$ is nonempty. Its new total in column $d \neq c$ is

$$\sum_{i \in S'} a'_{id} = s_d - a_{rd} + (s_c - 1)a_{rd} = s_d + (s_c - 2)a_{rd}.$$

If $s_d = 0$, then $a_{rd} = 0$ and the total is zero. Otherwise $s_d \geq 2$ and the total is at least two. Thus S' is a stopping set. □

Theorem 2.3 (Complete raw cancellation equals peeling; prior result)**.** *For every finite nonnegative integer matrix, the following are equivalent: there exists a complete positive singleton sequence; column-leaf peeling deletes every row; and there is no nonempty stopping set. Peeling's terminal row set is independent of queue order and is the greatest stopping set, with the empty set allowed as the greatest residual.*

Proof. A nonempty stopping set excludes a complete positive sequence by Lemma 2.2. If peeling stops with live rows, their column totals are zero or at least two, so these rows form a stopping set. Consequently the absence of stopping sets forces peeling to finish, and peeling itself is a positive singleton sequence.

Any stopping set in the original matrix survives every peeling move unchanged: a live column of total one cannot meet it. Hence every original stopping set is contained in the terminal row set. That terminal set is itself a stopping set when nonempty. It is therefore the greatest one and does not depend on the order of valid leaves. Equivalently, unions of stopping sets are stopping sets: a column meeting the union meets one of its stopping subsets, which already contributes at least two. □

Corollary 2.4 (Heredity under row deletion)**.** *Every row restriction of a completely peelable matrix is completely peelable.*

Proof. A stopping subset in the restricted matrix would be exactly the same stopping subset in the original matrix. Apply Theorem 2.3. □

Heredity is why a successful independent set of dual deletions can later be extended to a spanning tree. It does not say that an arbitrary primal column deletion preserves success; deleting columns can remove the only available leaves. The new geometric theorem repairs precisely that asymmetry.

2.1 Saturation and immutable-source replay

The map $x \mapsto \min\{2, x\}$ preserves whether a nonnegative sum or product is zero, one, or at least two. Therefore all pivot-admissibility and stopping-set decisions can use the three values $0, 1, 2^+$ without expanding large counts. In contrast, this saturation does not preserve parity: three becomes two. Geometric parity conditions must be proved on the uncapped source occurrences first.

During peeling, source coefficients never change. Keep, for each column, the number of remaining incident rows and the exclusive-or of their identifiers. If there is one owner and its source coefficient is one, the column is a leaf. Deleting that row updates each incident column once. With indexed adjacency lists this is $O(|R| + |C| + I)$ work and storage, where I is the number of nonzero incidences. Dictionary lookup, label encoding, and graph construction costs are separate implementation considerations.

A positive certificate lists distinct row/column pivot pairs. A checker can validate them directly against immutable source rows, or use the existing timestamp criterion: at the deletion time of row r, every other owner of its chosen column must already be gone, and the coefficient at r must be one. Successful production does not authorize a checker to trust an unauthenticated occurrence matrix.

2.2 Why signed homology is insufficient

A curve traversing a belt disc three times can have algebraic intersection one. Its raw count is still three, so it is not a geometric singleton in the current charts. Likewise, a determinant equal to one can certify a homological condition without exhibiting the disjoint attaching curves needed for handle cancellation. The incoming homological progress work can guide surface choices, but its output cannot be silently substituted for the raw matrix in this theorem. Later sections discuss a different, carefully scoped terminal use of verified group presentations.

3 Postponing the lower forest eliminates a search variable

The previous complete dual-forest search held the zero/one forest fixed. The previous weighted-tree proposal alternated additive optimizations of both forests. For the specific objective of *complete raw one/two cancellation*, however, a different argument eliminates the lower-forest search entirely. The producer should first find column leaves in the full source incidence matrix, and only then choose a zero/one forest that avoids its pivots. The actual geometric certificate still contracts this chosen forest *before* performing its one/two cancellations. Only discovery is postponed; no commutation of geometric moves is assumed.

3.1 The graph constraint that the matrix-only model forgets

Let $G = (V, E)$ be a finite multigraph, with loops and parallel edges allowed. Write $c(G)$ for its number of connected components. A spanning forest always spans each original component. Let R be a finite row set, and let J be a disjoint set of auxiliary columns. Consider a nonnegative integer matrix

$$N = (n_{rx})_{r\in R,\ x\in E\sqcup J}.$$

Assume that, for each row r, its restriction $(n_{re})_{e\in E}$ is the literal edge-occurrence vector of a closed walk in G. A union of closed walks also suffices. An empty restriction is permitted when a row has only auxiliary columns. Equivalently for the argument below, it suffices that the uncapped occurrence vector reduce modulo two to a graph cycle:

$$\sum_{e\ni v} n_{re} = 0 \pmod 2 \qquad (v \in V),$$

where a loop contributes twice. One must formulate this condition before saturation: replacing a coefficient three by two preserves singleton tests but not parity.

A *column-leaf move* chooses a live column x for which

$$\sum_{r\in R_{\text{live}}} n_{rx} = 1.$$

It deletes its unique incident row and deletes column x. Other entries are unchanged. All earlier pivot columns have zero occurrence in every later live row. Thus source entries remain immutable, and the existing sparse queue and independent timestamp checker apply unchanged.

Lemma 3.1 (A closed walk cannot traverse a bridge exactly once)**.** *If e is a bridge of a multigraph and γ is a closed walk, the number of traversals of e by γ is even. In particular, it is never one.*

Proof. Deleting e partitions its original component into two vertex sets. Every traversal of e changes the side occupied by the walk. The initial and final vertices agree, so the number of side changes is even. A loop is never a bridge. The same proof is the mod-two cut-cycle orthogonality identity and therefore applies to a union of closed walks. □

Theorem 3.2 (Pivot deletion preserves the source components)**.** *Run any finite column-leaf sequence on the full matrix N, and let P be its pivot columns that belong to E. Then $G - P$ has the same connected components as G. Consequently there is a spanning forest T of G with $T \cap P = \varnothing$.*

Proof. Order the moves chronologically. Auxiliary pivots do not remove graph edges. Suppose the next graph pivot is e, in row r, and let P_0 be the graph pivots of earlier moves. When an earlier column was a leaf, all rows that would survive that move had zero entry in that column. In particular $n_{rp} = 0$ for every $p \in P_0$. The closed walk for r therefore lies in $G - P_0$. The current pivot condition gives $n_{re} = 1$. By Lemma 3.1, e is not a bridge of $G - P_0$. Deleting it preserves every component. Induction proves the assertion. A spanning forest of $G - P$ now also spans G. □

In graphic-matroid language, P is independent in the cographic matroid: its complement contains a spanning forest. This observation is a consequence of closed attaching curves, not a property of arbitrary nonnegative matrices equipped with an unrelated graph.

Theorem 3.3 (Exact removal of the primal-forest choice)**.** *Under the hypotheses above, the following conditions are equivalent:*

1. *Full-matrix column-leaf peeling deletes every row of N.*
2. *There exists a spanning forest T of G such that column-leaf peeling deletes every row after columns T are discarded.*
3. *There exists a spanning forest T of G such that some sequence of raw positive-singleton cancellations deletes every row after columns T are discarded, with the usual nonnegative fill update.*

Whenever these conditions hold, a full-matrix peeling sequence followed by a spanning-forest computation in $G - P$ constructs T and its complete certificate. There is no enumeration of primal forests.

Proof. For $(1) \Rightarrow (2)$, choose the forest supplied by Theorem 3.2. No recorded pivot is in T. Deleting additional nonpivot columns does not change any pivot-column sum, so the identical row/column sequence is valid after discarding T. The implication $(2) \Rightarrow (3)$ is immediate: column leaves are zero-fill positive-singleton moves.

For $(3) \Rightarrow (2)$ use Theorem 2.3: a nonnegative matrix admits a complete positive-singleton sequence exactly when its column-leaf process deletes all rows. Finally, any column-leaf sequence using only columns outside T is also valid in the full matrix. Its pivot-column sums are identical, and the extra columns do not invalidate leaves. Hence $(2) \Rightarrow (1)$. □

Corollary 3.4 (A forest-independent refusal certificate)**.** *If full-matrix peeling leaves a nonempty stopping set $S \subseteq R$, then no primal forest permits complete raw positive-singleton elimination. The assertion concerns the specified geometric intersection model; it is not a certificate that the represented manifold is not a handlebody.*

Proof. Every full-matrix column sum on S is zero or at least two. Restricting the column set by deleting any forest preserves that property. Thus S remains a stopping set in every forest-complement matrix. The full-row cancellation theorem excludes a complete sequence in each such matrix. □

The stopping certificate needs only the nonempty list of source row identifiers. An independent checker reconstructs the uncancelled source occurrences, saturates each column total on that list at two, and rejects if any total is one. No forest search is needed to verify the universal quantifier in the corollary.

3.2 What is optimized when peeling stops

Let S be the terminal full-matrix column-leaf residual. For a forest T, let S_T be the terminal residual after discarding T. The greatest stopping-set property gives $S \subseteq S_T$ for every T. Moreover, the forest constructed from the complete *partial* full-matrix peeling sequence attains $S_T = S$: the recorded prefix is valid, and the remaining rows S still form a stopping set. Thus delayed forest discovery also finds the smallest possible terminal column-leaf residual over all primal forests.

This last statement must not be strengthened to arbitrary positive-singleton partial schedules. The matrix

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

has no column leaf, but a positive singleton pivot deletes one row and leaves the single entry two. Both rows can be realized combinatorially as the same closed walk ab on a graph with one vertex and two loops. The all-forest theorem optimizes *complete success*, and the partial

statement optimizes *column-leaf residuals*; it does not solve a maximum-length positive-fill cancellation problem.

3.3 A small strict example

Let G have two vertices and three parallel edges e_0, e_1, e_2, all oriented from the first vertex to the second. Take closed walks

$$\gamma_0 = e_0 e_2^{-1} e_1 e_2^{-1}, \qquad \gamma_1 = e_1 e_2^{-1}.$$

Their unsigned occurrence matrix is

$$N = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}.$$

The canonical forest $\{e_0\}$ leaves

$$\begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix},$$

which is a stopping set. Full-matrix peeling instead uses pivots (γ_0, e_0) and (γ_1, e_2). Their deletion leaves the spanning forest $\{e_1\}$; contracting this forest first gives the triangular matrix

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix},$$

with precisely the same valid cancellation sequence. This example is a closed-walk incidence construction, not a claim that arbitrary graph walks constitute authenticated manifold charts. Actual source examples are recorded below.

3.4 Source rows, annular rows, and the order of authentication

For a fixed source-authenticated two/three reduction, retain its surviving two-handle rows and all original zero/one handles. Each retained attaching cycle is literally a closed walk through these handles. The theorem therefore applies after any chosen legal dual forest. It eliminates the lower forest as a search variable for each dual choice; it does not decide which upper forest should be chosen.

For the finite bundle-boundary kernel, use the *whole* retained row set, including annular gluing rows, before choosing the source forest. The existing midpoint tracer `source_core_circle` checks that each annular core path closes in the original zero/one graph. The dedicated base columns are auxiliary columns J and are not members of the source forest. A source-column pivot still has a closed core walk through it exactly once, so Theorem 3.2 continues to apply. Omitting columns permanently incapable of providing a singleton must retain the already-proved lifting hypothesis of the finite kernel; the delayed-forest argument does not replace that hypothesis.

This extension cannot simply ignore annular rows while selecting a forest. A source-core leaf can cease to be a leaf after annular rows are added. The full finite boundary matrix must therefore be assembled first, and its source pivot columns determine the deferred forest.

The core verifier already validates an arbitrary supplied spanning forest against the authenticated source graph. It accepts the new producer's existing `normal-core-handlebody-v2` records unchanged. For the whole-boundary kernel we implement the new schema `bundle-boundary-kernel-v3`: its independent checker validates the supplied forest, reconstructs the post-forest matrix from source geometry, and checks the same cancellation and

census claims. Older schemas retain their canonical-forest meaning. In every case, the geometric replay contracts the forest before its one/two cancellations.

These certificates concern the supported fixed core or finite boundary matrix. Conservative returns elsewhere in the wrapper, the chosen dual forest, omitted-column lifting conditions, and construction of a bounded marked hierarchy successor remain separate obligations. No assertion about arbitrary group-word simplifications or commuting geometric moves is used.

3.5 Finite discovery cost

Let L be the number of literal source and annular edge occurrences, and let I be the number of nonzero saturated row/column incidences. Constructing the matrix costs $O(L)$ indexed operations, and the column queue uses $O(I + |R| + |E| + |J|)$ indexed work. A depth-first or breadth-first traversal of $G - P$ finds a spanning forest in $O(|V| + |E|)$ work. Thus the abstract discovery algorithm has a linear bound in these finite explicit parameters and uses linear space.

The shipped implementation reuses the existing deterministic disjoint-set helper, with path halving and minimum-label linking, to retain the original component identifiers. It does not use union by rank or size; we therefore do not attach the usual inverse-Ackermann bound to that helper. Sorting, component-census loops, Python integer/dictionary operations, source preparation, and certificate replay also have their own costs. The linear indexed theorem is not a claim that the entire native producer is linear in its encoded input. Nothing in deferred discovery requires expanding a family represented only by an omitted genus counter.

Native core and whole-cut measurements are reported separately in Section 8.

4 A geometric family with exponentially many lower forests

The small closed-walk example extends to a family in which the candidate forest space is exponential, although delayed discovery is linear in the input size. This does not establish a lower bound on all competing algorithms: the particular family also admits a direct block decomposition. Its purpose is to prove a strict, scalable difference between a canonical forest policy, unpruned forest enumeration, and the new complete strategy.

Theorem 4.1 (Theta-block separation family)**.** *For every $h \geq 1$, there is a connected graph G_h with $h + 1$ vertices and $3h$ edges, and $2h$ specified closed attaching walks, with the following properties.*

1. *G_h has exactly 3^h spanning trees. Exactly 2^h of them permit complete raw positive-singleton cancellation of all $2h$ rows.*
2. *The source-order canonical forest leaves every row in a stopping set. Full-matrix peeling deletes all rows, using $5h$ nonzero incidences, and then obtains a disjoint spanning forest in $O(h)$ indexed work.*
3. *These unsigned attaching incidences admit a genuine geometric handle realization of a three-ball. This existence statement does not assert that the repository's normal-cut front end produces that handle structure.*

Proof. Take vertices $v_0, v_1, \ldots, v_h$. For each i, join v_0 to v_i by three parallel edges a_i, b_i, c_i, in this source order. The two closed walks in block i have words

$$a_i c_i^{-1} b_i c_i^{-1}, \qquad b_i c_i^{-1}.$$

Their raw incidence block, with columns a_i, b_i, c_i, is

$$N_i = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}.$$

There are no incidences between blocks. Every spanning tree must contain exactly one edge from each triple, so there are 3^h trees.

Choosing a_i in a block leaves $\left(\begin{smallmatrix} 1 & 2 \\ 1 & 1 \end{smallmatrix}\right)$, a nonempty stopping set. No operations in other blocks can change that obstruction. Choosing b_i or c_i leaves respectively

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

each completely column-peelable. Thus success is equivalent to choosing one of b_i, c_i independently for every block, giving exactly 2^h successful trees. The canonical tree chooses every a_i, so its entire $2h$-row matrix is blocked. In the full matrix, column a_i first removes the first row of block i, after which b_i or c_i removes the second. There are $6h$ literal occurrences and $5h$ nonzero incidences, and each is processed only a bounded number of times.

For geometric realization, begin with two zero-handles and three one-handles a_i, b_i, c_i joining them. A zero-handle boundary with its three one-handle feet removed is a pair of pants. Choose two disjoint seams joining the a_i foot to the c_i foot and the b_i foot to the c_i foot. Choose compatible seams at the other zero-handle. Distinct product strips through the one-handles join the seams into two disjoint simple attaching curves x_i, y_i whose literal passages are $a_i c_i^{-1}$ and $b_i c_i^{-1}$. Contracting c_i gives a single zero-handle, and the two curves cancel the remaining one-handles, so attaching two-handles along this system gives a ball.

Slide the two-handle attached along x_i over the one attached along y_i. Realize the slide by an embedded band in the boundary of the central zero-handle, with interior disjoint from both curves and every one-handle foot, and use a parallel copy of y_i. Such a band exists in the complement of the two chosen seams. The slid curve and the original y_i remain disjoint and simple. The based word can depend on orientation and basepoint conventions; the theorem asserts realization of the unsigned incidence matrix, for which either orientation of the parallel copy has the same count. Since the band itself stays in the zero-handle, its raw passages are precisely the sum of the passages of x_i and y_i, namely $(1, 1, 2)$. The second row remains $(0, 1, 1)$. A geometric two-handle slide preserves the three-manifold, so this is still a handle decomposition of a ball.

Place h copies in disjoint regions of one central zero-handle, with separate outer zero-handles. Equivalently, take boundary connected sums of the constructed balls and merge their central zero-handles. The result is a three-ball with exactly the graph G_h and the stated unsigned block incidence matrix. The proof is an explicit handle construction; converting it to the particular authenticated normal-parallelity source schema remains separate implementation work. □

Proposition 4.2 (Cost of a specified unpruned enumeration)**.** *For the family above, enumerate spanning trees in lexicographic order $(s_1, \dots, s_h) \in \{0, 1, 2\}^h$, with choices a_i, b_i, c_i respectively, and run the column-leaf test separately on each tree. The first success occurs after*

$$\frac{3^h + 1}{2}$$

tested trees.

Proof. The lexicographically first successful vector is $(1, \dots, 1)$. Its zero-based ternary index is $\sum_{j=0}^{h-1} 3^j = (3^h - 1)/2$. Including this final successful trial gives the displayed count. □

A block-aware solver does not need this enumeration, and neither does the deferred-forest theorem. The exact count concerns the stated baseline, not an intrinsic exponential lower bound. The saved benchmark driver reports all raw timings for this baseline, for deferred discovery on the same family, and for both native core policies on the common success subset. Source geometry preparation, dual reduction, and certificate replay are excluded from those kernel timings and must not be conflated with end-to-end recognition time.

5 Fixed-parameter forest selection beyond two-row columns

The two-row theorem of [9] reduces a restricted dual-forest choice to directed reachability and independent representatives in a graphic matroid. The remaining original native column shape is $(1, 1, 1)$: deleting two of its rows allows the third to peel. Such a rule has two premises and cannot be represented by ordinary directed reachability alone. This section gives an exact reduction when the rows participating in these exceptional rules form a small interface. The parameter is independent of the number of interior dual vertices.

5.1 The finite problem and its parameter

Let D be a connected finite undirected multigraph, with a distinguished root and vertex set V. Let R be a finite row set. Each row $r \in R$ labels either an edge of D or an ineligible record that cannot be selected in a forest. Eligible row records are in bijection with the edges of D, so distinct parallel edges still have distinct row labels. Loops and parallel edges are allowed. Write

$$A = (a_{rc}) \in \mathbb{N}^{R \times C}$$

for a nonnegative integral matrix, supplied by sparse occurrence lists. In the geometric application, a row is an attaching two-handle and a column is a retained one-handle cocore. That interpretation requires the separate source-authentication and geometric cancellation theorems; the finite operator below does not establish it from the matrix alone.

A spanning tree $T \subseteq R$ deletes its rows. A *unit-column peel* on a current live row set L removes a row $r \in L$ when some column c satisfies

$$\sum_{s \in L} a_{sc} = a_{rc} = 1.$$

Its column may also be deleted, since it has no other live occurrence. The decision problem is whether some spanning tree leaves a row restriction that can be completely peeled. By the raw-cancellation theorem of [9], this is also the exact question for complete positive raw cancellation in the specified occurrence model. A negative answer concerns that model and its fixed geometric charts.

Call a column *active* if it has an entry equal to one. An inactive column cannot supply a peel after any row deletion, so it can be ignored for this decision. For a column c, let

$$W_c = \{r \in R : a_{rc} > 0\}, \qquad E = \{c \in C : c \text{ is active and } |W_c| > 2\}.$$

The *exceptional interface* and its size are

$$U = \bigcup_{c \in E} W_c, \qquad u = |U|. \tag{1}$$

Inert columns with large supports do not contribute to U. Conversely, the support of an exceptional active column contributes *all* its rows, including rows whose own entry in that column exceeds one.

Throughout this section, N denotes the encoded input size, including graph records, sparse incidences, and the binary encodings of counts. Counts can be capped at two for the zero/one decisions used here. Such capping must not be used to justify a separately asserted parity property of the original geometric chain complex.

5.2 Baseline reachability and mandatory source components

Temporarily omit the columns of E. Build a directed graph P on R from the remaining active columns. A singleton column with entry one at r makes r an automatic seed; let Z be the set of these seeds. A column supported on $\{r, s\}$ gives the arc $s \to r$ if $a_{rc} = 1$, and the arc $r \to s$ if $a_{sc} = 1$. No restriction is imposed on the other positive entry. Reachability includes paths of length zero.

Lemma 5.1 (Exact baseline activation)**.** *If a row set Q is deleted initially, the rows deleted or peeled using only the baseline columns are exactly*

$$\mathrm{Reach}_P(Z \cup Q).$$

For an exceptional column c, its additional exact activation rules are

$$W_c \setminus \{r\} \implies r \qquad (r \in W_c,\ a_{rc} = 1). \tag{2}$$

Consequently closure under baseline reachability and these rules is exactly the set of initially deleted or subsequently peelable rows.

Proof. An automatic seed can peel without another row deletion. For a two-row column, deleting the other supported row makes its unit entry a legal pivot. Thus every directed path gives a valid sequence of possible activations. Conversely, each baseline peel is either an automatic seed or follows an arc whose tail is already activated. Induction on a peel trace proves the first assertion, recovering the reachability model of [9].

For an exceptional column, a live row r can peel in that column exactly when its entry is one and every other supported row has been deleted. These are precisely the premises of (2). Interleaving these rules with baseline activations proves exactness for the full matrix. Continuing to retain a column in this logical description after its pivot is harmless: at that pivot it had no other live owner, and therefore cannot later remove a different live row. □

Let $S_1, \ldots, S_q$ be the source strongly connected components of P that contain no automatic seed and are disjoint from U. They are pairwise disjoint, and their number need not be small. Put

$$M = \bigcup_{i=1}^{q} S_i, \qquad B = \mathrm{Reach}_P(Z \cup M), \qquad B_U = B \cap U. \tag{3}$$

The definition of B treats all rows of M as activated for convenience; the algorithm will actually require only one selected forest row from each S_i.

Lemma 5.2 (Mandatory groups and an interface completion criterion)**.** *Every initially deleted row set that permits full peeling must meet each S_i. A single initially deleted row in S_i activates the entire component by baseline reachability. If Q meets all these groups, then full activation of R is equivalent to activation of every row in U.*

Proof. The component S_i has no incoming arc in the condensation graph, no automatic seed, and no row in the support of an exceptional rule. Hence neither baseline propagation from outside nor an exceptional activation can create its first activated row. It must contain an

initially deleted row. Strong connectivity then activates all its rows and every baseline descendant.

For the last assertion, activation of all of R plainly activates U. Conversely, every source strongly connected component either contains an automatic seed, is one of the groups S_i, or intersects U. In the third case activation of all of U supplies a row of that component; in the first two cases a row is already supplied by Z or Q. Every source component is therefore activated in full. Each component of a finite directed acyclic condensation graph is reachable from a source, so all of R becomes activated. □

5.3 Compressing all nonlinear interaction to the interface

For any row $r \in R$, define its reachability signature

$$\sigma(r) = \mathrm{Reach}_P(\{r\}) \cap U. \tag{4}$$

On subsets of U, define Hcl to be the least closure under the following finite rules:

1. $v \Longrightarrow w$ whenever $v, w \in U$ and w is reachable from v in P;
2. every implication in (2).

All premises and conclusions in the second family lie in U by its definition. Thus this closure can be computed without retaining the non-interface vertices as dynamic states.

Lemma 5.3 (Exact compressed Horn closure)**.** *Let $Q \subseteq R$ meet every mandatory group S_i. The interface rows eventually activated from the initial deletion of Q are exactly*

$$\mathrm{Hcl}\left(B_U \cup \bigcup_{r \in Q \setminus M} \sigma(r)\right). \tag{5}$$

In particular, $A[R \setminus Q, C]$ is completely peelable if and only if the set in (5) is U.

Proof. Each mandatory group has a selected row, so all of B can be activated by the baseline rules. Every $\sigma(r)$ in the displayed expression is reachable from an actual initially deleted row. Every interface reachability rule abbreviates a valid baseline path, and every exceptional rule is exact by Lemma 5.1. The displayed closure is therefore contained in the actual activated interface.

For the reverse inclusion, consider an activation trace and separate its baseline propagation from its exceptional rule firings. Before any exceptional firing, each activated interface row is reached from Z or Q. Paths from Z or $Q \cap M$ are covered by B_U; paths from the other rows of Q are covered by their signatures. If an exceptional rule fires, its head belongs to U, and all subsequent baseline propagation from this head to interface rows is represented by interface reachability rules. The exceptional implication itself is retained without any change of premises. Induction on the exceptional firings proves that every actual interface activation occurs in (5). The final equivalence follows from Lemma 5.2. □

The compression is algorithmic as well as logical. Reverse reachability from each $v \in U$ computes all signatures: a row r receives the target v exactly when r reaches v in P. Forward reachability from the interface rows computes the interface-to-interface rules. These computations use a polynomial number of graph operations. Repeatedly scanning the finite rule list until no target is added computes Hcl; there are at most u strict additions.

5.4 Partial partitions and independent representatives

A *partial partition* of U consists of a subset $U_0 \subseteq U$ and a partition

$$\Pi = \{B_1, \ldots, B_\ell\}$$

of U_0 into nonempty blocks. The empty subset has the empty partition. The interpretation is that U_0 is a guessed collection of interface targets activated directly by selected tree rows through baseline paths; one additional tree row is to supply each block. This is a witness assignment, not an assertion that all possible tree seeds have already been identified.

For each partial partition, first test

$$\mathrm{Hcl}(B_U \cup U_0) = U. \tag{6}$$

If this fails, discard the partition. Let R_{el} consist of the rows labelling eligible nonloops of D. Otherwise define the candidate group for its jth block by

$$G_j = \{r \in R_{\mathrm{el}} \setminus M : B_j \subseteq \sigma(r)\}. \tag{7}$$

Seek distinct representatives, one from each member of the group list

$$S_1, \ldots, S_q, G_1, \ldots, G_\ell, \tag{8}$$

whose graph edges form a forest. Ineligible rows and loops can also be removed from the mandatory candidate groups when performing this test. An empty group makes that particular test fail.

Lemma 5.4 (Overlapping groups reduce to graphic/partition intersection)**.** *The independent-representatives problem for* (8) *is decidable in polynomial time, even though the block groups G_j may overlap. It requires at most $(\ell + 1)|R|$ graph-edge copies.*

Proof. Make one edge copy (r, H) for each permitted membership of row r in a group H from (8). Give this copy the same endpoints as r and color it by H. Copies of a single original edge are parallel in the resulting multigraph. Two such copies cannot both belong to a forest, since they form a two-edge cycle. Graphic independence therefore enforces both distinct original representatives and independence of the original edge set. The partition matroid that allows at most one copy of each color enforces the group constraints. A common independent set of size $q + \ell$ exists exactly when the desired representatives exist.

This is the standard graphic/partition matroid-intersection problem, with polynomial independence tests and a polynomial augmenting-path algorithm [2, 14]. It is also Rado's independent-transversal formulation, already used in [9]; no new matroid-intersection theorem is needed. The mandatory groups are disjoint and contribute at most $|R|$ memberships in total. Every other row contributes at most one copy per block group, hence at most $\ell|R|$ additional memberships. This proves the stated size bound. □

The exclusion of M from the block candidates loses no useful initial activation. Once the mandatory groups have their representatives, all signatures of rows in M are already included in the baseline contribution B_U. The overlapping candidate groups outside M must nevertheless be handled explicitly; treating them as disjoint colors on the original edge set would be incorrect.

Theorem 5.5 (Exact bounded-interface forest selection)**.** *The following finite procedure decides whether some spanning tree of D leaves A completely peelable:*

1. *Construct the baseline graph, automatic seeds, mandatory groups, signatures, and interface closure.*
2. *Enumerate all partial partitions of U. Retain only those satisfying* (6).
3. *For each retained partition, solve the independent-representatives problem* (8).
4. *If representatives form a forest Q, extend Q to a spanning tree T of D, recompute a complete peel order on $A[R \setminus T, C]$, and return T and that order.*
5. *If all partial partitions have failed, report that no spanning tree in the specified finite class permits full peeling.*

The algorithm makes at most $\mathrm{Bell}(u+1)$ *polynomial independent-representatives calls. Its bit complexity is*

$$2^{O(u \log(u+2))}\,\mathrm{poly}(N). \tag{9}$$

Proof. Suppose the algorithm finds representatives forming Q. Each mandatory group is met. Each block B_j is contained in the actual signature of its representative, so the actual initial interface activation contains $B_U \cup U_0$. Equation (6) and Lemma 5.3 show that deleting Q permits activation of every row. Thus $A[R \setminus Q, C]$ is completely peelable. Every forest in a connected graph extends to a spanning tree. Deleting additional rows preserves peelability: restrict an existing peel order to the rows still present; at each remaining pivot its own occurrence is one and the other live occurrences remain absent. Hence the extension T also succeeds. Recomputing the final trace provides an ordinary tree-and-peeling certificate on the actual chosen complement. This proves soundness.

For completeness, suppose a successful spanning tree T exists. Lemma 5.2 shows that it meets each S_i; choose one of its rows from each group. Define

$$U_0 = \bigcup_{r \in T \setminus M} \sigma(r).$$

Since T succeeds, Lemma 5.3 implies (6). For each $v \in U_0$, choose an actual witnessing row $r(v) \in T \setminus M$ with $v \in \sigma(r(v))$. Partition U_0 by equality of these witnesses. The resulting nonempty blocks have distinct representatives in $T \setminus M$, and each block is contained in the signature of its representative. Together with the chosen mandatory representatives, they form a subset of T and are therefore independent. The algorithm enumerates this partial partition, and its representatives test succeeds. This argument proves why at most u extra representatives suffice; it does not assume that arbitrary tree edges can be replaced by source rows while preserving independence.

For the count, adjoin one distinguished element $*$ to U. A partition of $U \cup \{*\}$ corresponds bijectively to a partial partition of U: the block containing $*$ records the omitted elements, and its other blocks are the blocks of the partial partition. There are exactly $\mathrm{Bell}(u+1)$ objects. Label the elements of U by $1, \ldots, u$ and give $*$ the label zero. A partition is determined by assigning each ordinary element the least label in its block. Thus, with the empty-interface case interpreted separately,

$$\mathrm{Bell}(u+1) \le (u+1)^u.$$

Every block list has $\ell \le u$, so Lemma 5.4 gives a graph of polynomial size for each call. Candidate membership uses ordinary subset tests on u-element signatures. Baseline reachability, interface closure, spanning-tree extension, and the final peel require polynomial time in the encoded input. Integer input reading and capping are polynomial in the binary bit lengths. Combining these observations yields (9). □

Remark 5.6 (The empty interface and a harmless implementation refinement). If $u = 0$, there is one partial partition, its block list is empty, and the procedure reduces to the prior polynomial

two-row theorem. More generally, one may first compute $U_{\text{base}} = \mathrm{Hcl}(B_U)$ and enumerate partial partitions only of $U \setminus U_{\text{base}}$. Targets already activated by the mandatory groups and automatic seeds need no additional witness. Any successful-tree witness assignment can simply omit those targets, so completeness is preserved. This can decrease the actual enumeration substantially; the bound in Theorem 5.5 remains valid without relying on it.

5.5 Native specialization and certificate semantics

Proposition 5.7 (Native triple-column specialization)**.** *Suppose the matrix is the original source-authenticated native core occurrence matrix for which each unsaturated column total is at most three. Let k be the number of its active columns supported on more than two rows. Then each such column has shape $(1, 1, 1)$ and*

$$u \leq 3k.$$

The finite forest selector therefore has complexity

$$2^{O(k \log(k+2))}\,\mathrm{poly}(N). \tag{10}$$

Proof. A column supported on at least three rows has at least three positive occurrences. The total bound of three forces exactly three supported rows and an entry of one at each. Its support contributes at most three rows to the union defining U. Taking the union over the k exceptional columns proves $u \leq 3k$, and substituting into (9) proves the complexity statement. The geometric three-occurrence premise is the prior native cocore theorem of [9], not a consequence of the finite matrix algorithm. □

This specialization applies to the original native source-core matrix and to row or column deletions that preserve its occurrence interpretation. Adding annular gluing relations or arbitrary boundary rows can increase a column's support beyond three. Such enlarged inputs remain covered by Theorem 5.5 with their actual parameter u; they are not automatically covered by the estimate $u \leq 3k$. Similarly, transporting additional markings through a chosen dual tree requires the applicable geometric port-preservation theorem.

The parameterized result places no restriction on $|V| - 1$, the number of tree edges to be chosen. It gives a local quasi-polynomial regime whenever the maintained encoded input is polynomial in a diagram parameter n and $u \log(u + 2)$ is bounded by a fixed power of $\log n$. In particular, $u \log(u + 2) = O((\log n)^2)$ would give a local $n^{O(\log n)}$ bound. No such interface bound for every state of a complete unknot-recognition hierarchy is proved here, and the cost of constructing or branching over those states remains a separate problem.

Remark 5.8 (Positive replay, exhaustive failure, and budgets). A positive output contains the final spanning tree and a complete unit-column peel trace. Their finite correctness can be checked directly without replaying the partial-partition search or trusting a matroid-intersection optimality claim. The enclosing geometric verifier must still authenticate the rows, columns, dual endpoints, and topological meaning of the cancellations.

An exhaustive negative result means that no tree works in this fixed finite class. The current implementation does not package the entire exhaustion as a succinct independent negative certificate. In particular, a rank defect for one partial partition is only a rejection of that partition; every possibility must be excluded before the complete finite-class negative is justified. Interrupting the enumeration because of a work allowance, partition allowance, or implementation limit must instead return an inconclusive status. None of these negative or interrupted outcomes is a certificate that the underlying knot is nontrivial or that an authenticated core is not a handlebody.

5.6 Why triangle-cycle parity does not remove all tree choice

The native parity condition constrains the odd support of each source column to be an even-degree subgraph of the dual graph. For a unit triple, one allowed form is a triangle. It is tempting to conclude that such short cycle relations make all spanning trees equivalent for raw peeling. The following finite witness rules out that inference from sparsity and parity alone.

Proposition 5.9 (An abstract triangle-column sensitivity witness)**.** *There is a connected simple graph D on nine vertices with 25 edges and a Boolean matrix with those edges as rows and 17 columns such that:*

1. *each column is supported on exactly the three edges of a triangle of D;*
2. *one spanning tree leaves a completely peelable row restriction;*
3. *another spanning tree leaves a row restriction admitting no complete positive raw cancellation.*

Thus the three-occurrence bound and dual-cycle parity condition together do not imply dual-tree invariance for the abstract finite problem. No native triangulated realization of this example is asserted.

Proof. Appendix B and the supplied file `triangle_tree_sensitivity.json` give the complete graph, all triangle-column supports, and two explicit spanning trees. Each tree has eight edges. For the successful tree, the file lists 17 unit-column peels removing all 17 surviving rows.

For the second tree, the recorded eight peels leave the nonempty row set

$$\{0, 4, 6, 7, 13, 17, 21, 22, 23\}.$$

Its column totals, in the supplied column order, are

$$(2, 3, 0, 0, 2, 0, 2, 2, 0, 0, 2, 0, 0, 2, 0, 2, 2).$$

No total is one, so this is a stopping set. The stopping-set persistence theorem of [9] excludes every complete positive raw cancellation schedule, including schedules with positive fill. The failure is therefore independent of the particular displayed peel order.

All finite assertions are checked by the supplied standard-library script `check_triangle_witness.py`, which imports no selector, peeling backend, or topology library. It checks that every column consists of three distinct edges with degree two at exactly three graph vertices, that each proposed tree is connected and has eight edges, that every recorded pivot has live column total one, and that the final residual has the displayed stopping totals. A triangle has even degree at every vertex and exactly three occurrences, establishing both claimed necessary compatibility conditions. □

The witness is an explicit finite counterexample to an abstract invariance claim. It does not establish how often dual choice matters on geometrically authenticated sources after other normalization improvements, nor does it exclude a stronger invariance theorem for a more restricted native family. Its role is to keep the distinction between the proved local parity constraints and a full geometric realization theorem explicit.

6 Combining the two forests, and a linear-algebra extension

6.1 An exact coupled finite problem

Suppose an authenticated handle model supplies a primal graph $G = (V, E)$, a connected rooted dual graph D, and an occurrence matrix $N \in \mathbb{N}^{R \times (E \sqcup J)}$. Each row restricts to a

closed walk in G. Eligible rows are in bijection with the dual edges; other rows may be ineligible for top-handle deletion. Let $\mathcal{L}$ be the total bit length of the graphs, matrix, and row/edge correspondence. Assume every dual spanning tree is a legal specified two/three-handle reduction that deletes precisely its selected rows, leaves the retained occurrence data unchanged, and preserves the exposed ports required by the subsequent operation. These are geometric hypotheses on the model, not automatic properties of a pair of graphs. Also assume that every chosen primal spanning forest has the authenticated forest-first contraction used in Section 3: it deletes those columns and preserves the retained attaching charts and required port interpretation. This lower contraction contract is part of the coupled geometric application.

Corollary 6.1 (Coupled forests without primal enumeration)**.** *Under these hypotheses, there exist a legal dual tree and a primal spanning forest permitting complete raw positive cancellation if and only if there exists a dual tree whose deleted-row complement is completely column-peelable on* all *columns $E \sqcup J$. The latter problem is decidable in*

$$2^{O(u \log(u+2))}\,\mathrm{poly}(\mathcal{L}),$$

where u is the exceptional-interface size of the full pre-deletion matrix. A successful decision constructs both trees and a complete raw cancellation sequence.

Proof. Fix a legal dual tree T_D. Every retained row still has its original closed-walk interpretation, so Theorem 3.3 applies to the row restriction $R \setminus T_D$. It gives the stated equivalence for this particular tree. Quantifying over the dual trees proves the first assertion. Apply Theorem 5.5 to the full matrix and dual graph; after its successful tree is found, peel the retained rows and choose a primal forest avoiding the graph pivots. The extra finite work is polynomial and introduces no new search factor. □

For original native source-core columns only, the prior bound $u \le 3k$ applies when k counts exceptional columns in this full input. It must not be estimated only after a favorable but unknown dual tree is chosen: that would assume the answer in the parameter bound. A separate parameterized algorithm would be needed if the parameter were instead the minimum exceptional size over all dual choices.

The delivered native patch uses the existing breadth-first dual policy and the new primal theorem. The delivered exceptional solver is a separate finite operator. Corollary 6.1 justifies a future source adapter under the stated contracts; it is not a claim that the current whole-cut wrapper already implements exhaustive coupled selection. In particular, no extra native successes are attributed to the standalone dual solver in our result table.

6.2 Deferred bases in a represented binary matroid

The connectivity argument is a special case of a useful algebraic fact. It also identifies exactly what would be lost by replacing a forest with a basis of a represented binary matroid.

Theorem 6.2 (Deferred column basis)**.** *Let B be a matrix over $\mathbb{F}_2$ with columns indexed by E, and let $N \in \mathbb{N}^{R \times (E \sqcup J)}$. Assume that each uncapped row vector on E lies modulo two in the kernel of B:*

$$B\,(n_{re} \bmod 2)_{e \in E} = 0 \qquad (r \in R).$$

If a column-leaf sequence in N uses E-indexed pivot set $P \subseteq E$, then

$$\mathrm{span}(B_{E \setminus P}) = \mathrm{span}(B_E).$$

Consequently $E \setminus P$ contains a column basis K of B. Moreover, full-matrix peeling succeeds if and only if some column basis K can be discarded while leaving a matrix admitting complete raw positive singleton cancellation.

Proof. Before a pivot $e \in E$ in row r, every earlier pivot column has coefficient zero in r. Its coefficient at e is one. The kernel identity expresses B_e as a sum of the other currently retained columns with coefficients $n_{rf} \bmod 2$. Deleting e therefore preserves the span. Auxiliary pivots change none of the columns of B. Induction proves the displayed equality, so a basis can be selected from the complement of all pivots.

If full peeling succeeds, choose this basis K. The recorded sequence uses no column of K and remains valid after those columns are discarded. Conversely, complete raw positive elimination after a basis deletion implies complete peeling by Theorem 2.3, and the same pivot-column sums make that peeling sequence valid with the deleted basis columns restored. This proves the equivalence. □

Take B to be the mod-two vertex/edge incidence matrix of a graph. Its column bases are exactly spanning forests, and the theorem recovers the primal result. More generally, a cellular boundary relation $\partial_k \partial_{k+1} = 0$ can supply the kernel hypothesis for a represented matroid, provided that the raw incidences have the claimed mod-two interpretation. This gives an algebraic normalization principle in other chain complexes.

An arbitrary column basis in a higher boundary matrix need not be geometrically collapsible. Therefore this generalization does not authorize higher-dimensional handle cancellation, replacement of a marked disk system, or compression of a manifold from homology alone. The special advantage of dimension one is that a graph basis is an forest with contractible components. Proving a similarly useful geometric realization criterion in another setting is a research problem.

6.3 Stopping sets as reusable restricted certificates

The universal primal obstruction is unusually cheap to store: a nonempty list of source row identifiers suffices. Source replay checks that every column total on those rows avoids one. Such a witness can be reused across all primal forest policies on the same retained rows and charts. It invalidates neither a different dual deletion nor a different geometric presentation. A cache must therefore bind the witness to the authenticated source, retained-row set, matrix version, and occurrence convention. Using an obstruction from an earlier hierarchy state after changing those data would be unsound.

The exceptional solver's exhaustive negative has a different status. Its search proves failure for every allowed dual tree in one finite instance, but the current implementation provides no compact independently replayable certificate of the entire exhaustion. These two kinds of refusal should have separate names in an eventual API.

7 Implementation and independent certificate replay

7.1 A small, opt-in native change

The patch changes four production modules and adds two test modules under `Topology/UnknotRecognition/fast`. Existing defaults remain canonical. The optional argument

```
forest_strategy='deferred'
```

is accepted by the core producer, finite bundle-boundary producer, and whole-cut wrapper. Unknown strategy values raise `ValueError`. The argument is propagated through the supported core, augmented-core, and finite-kernel fallback calls.

In the core producer, deferred discovery builds the full occurrence matrix, runs the existing column queue, and chooses a spanning forest avoiding the recorded pivot columns. It checks that the original component partition is unchanged. If peeling stops, the result remains `INCONCLUSIVE`, with reason `all_primal_forests_raw_obstruction` and the remaining two-handle identifiers. On success it emits the existing `normal-core-handlebody-v2` schema. The independent core checker already accepts a general supplied forest, so its code is unchanged.

The logical producer is short:

```
rows = source_occurrences_without_a_primal_forest()
trace, residual = column_leaf_peeling(rows)
if residual:
    return INCONCLUSIVE, restricted_stopping_witness(residual)
pivots = source_graph_columns_used_by(trace)
forest = spanning_forest(source_graph_without(pivots))
assert original_components_are_preserved(forest)
return certificate(forest_first=True, forest=forest, trace=trace)
```

This pseudocode describes one supported finite matrix. The full wrapper still has conservative branches, port conditions, and bundle lifting requirements outside this local argument.

7.2 Why the whole-boundary change needs a new schema

Core success is insufficient when a whole cut is reconstructed by gluing parallelity bundles. The attaching annuli add rows and may destroy leaves that were available in the isolated core. Deferred boundary discovery therefore first includes every supported annular row, using the authenticated closed core paths and dedicated auxiliary base columns. It then peels this full finite matrix and chooses the source forest avoiding its source pivots.

The old boundary schemas reconstruct a canonical forest. Reinterpreting them as permitting an arbitrary supplied forest would change an existing certificate's meaning. The patch instead introduces `bundle-boundary-kernel-v3`; earlier schemas retain their exact validation policy. The outer whole-cut certificate schemas need not change because their kernel field already dispatches by kernel schema.

The v3 checker independently decodes and validates every supplied forest edge. It rejects booleans masquerading as integer identifiers, malformed hexadecimal strings, negative or out-of-range handles, duplicates, loops, cycles, and a failure to span the original graph components. It uses authenticated endpoints to reconstruct the component partition, rather than calling the producer's forest routine. It then reconstructs raw post-forest source and annular occurrences, validates the cancellation trace, and recomputes the component census, base families, and bridge information.

The component identifiers remain the minimum source vertex labels, so choosing another forest does not silently rename component ownership. The tree is removed from the surviving-column census after discovery; forgetting that removal would preserve a valid cancellation trace but give an incorrect genus. Both steps are covered by direct kernel and whole-cut replay tests.

7.3 Checking independence and hostile certificates

The new tests monkeypatch producer entry points, the producer's forest helper, the peeling routine, and a census helper to raise if called while verifying a saved positive packet. Verification still succeeds. This tests the intended separation at the modified interfaces; it is not a claim that source preparation and all geometry code are formally independent implementations.

Mutation cases remove a tree edge, duplicate an edge, insert invalid labels, remove a required cancellation, select a forest edge as a cancellation pivot, or relabel a v3 packet as v2. All are rejected. Further tests exercise callbacks, exhausted operation limits, source relabelings, an exact literal oracle comparison, and a normal-coordinate scale factor 2^{128}. The scaling test checks that binary multiplicity does not force literal normal-sheet expansion in the changed path; it does not establish an asymptotic runtime bound by measurement alone.

7.4 The standalone exceptional solver

The research operator `exceptional_forest.py` depends only on Python's standard library. It uses the attributed, unchanged `degree_two_backend.py` from the incoming dual-normalization package for validation, strongly connected components, classical matroid intersection, forest extension, and final peeling. Its new code computes the exceptional interface and signatures, performs Horn closure, enumerates partial partitions, and clones overlapping candidate memberships.

The input has one optional dual-edge record per matrix row. A null record is ineligible for the selected tree, and loops cannot be selected. The root is vertex zero. Sparse row entries retain exact positive integer counts on input, although the solver caps them at two for singleton decisions. The three statuses mean:

`SUCCESS`
: A final spanning tree and complete peel sequence have been constructed. A separate finite checker can verify both.

`NO_FOREST`
: The complete finite search has excluded every tree in the specified graph/matrix problem. This is not a topological negative certificate.

`INCONCLUSIVE`
: A work/partition budget or implementation limit interrupted the search. Exhaustion of a resource is never treated as exhaustive mathematical failure.

No native extraction adapter for this operator is included. Such an adapter should use the existing authenticated dual endpoints and full raw source occurrences, feed any successful tree into the legal top reduction, and obtain ordinary native replay before claiming a new topological success. The finite solver's tests deliberately do not assume geometric realizability of random matrices.

7.5 Integration policy

The direct patch preserves canonical behavior by default. This is a practical decision: the stronger deferred test adds work on many inputs that the canonical policy already settles. A later adaptive policy could try canonical peeling first and defer only when needed, but should reuse authenticated rows and avoid rebuilding expensive geometry. We do not include that additional optimization without measuring its full cost.

The artifact contains both a patch against the pinned commit and an overlay of the six

changed files. Apply one representation, not both. It also contains source hashes, the original baseline versions of the four production modules for comparison, exact new packets, and the audit drivers. Existing source validators, the normal-orbit construction, dual breadth-first policy, and the outer knot verdict logic retain their own contracts; local completeness must not be read as a claim about every branch of the whole wrapper.

8 Reproducible experiments and their limits

All native comparisons start with the same source triangulation and normal-coordinate vector. Source files are hashed before and after the whole-cut audit. Every positive native output is replayed by the existing outer verifier, including the new kernel schema where present. Raw JSON, complete new packets, commands, logs, and independent finite checkers accompany the article. The counts below refer to this fixed corpus, not a random sample of all knots or all normal surfaces.

8.1 Native coverage: distinguish the core from the whole cut

Operation	Corpus	Cases	Canonical	Deferred
Core certificate	Historical	106	100	106
Core certificate	Additional interior cases	14	10	12
	Total	120	110	118
Whole-cut census	Historical	106	92	99
Whole-cut census	Additional interior cases	14	10	12
	Total	120	102	111

Table 1: Complete source-authenticated certificates. All previous positive cases remain positive. An inconclusive result is not a negative manifold or knot verdict.

After the existing dual breadth-first reduction, every historical source core now peels, and the additional interior corpus has two remaining core obstructions. Their greatest stopping residuals contain ten rows for `nested_1_4_6/interior_links` and four for `interior_loop/interior_links`. The independent stopping checker validates each residual on reconstructed core occurrences. The universal quantifier is over primal forests for that fixed retained matrix; neither another dual choice nor another move class is excluded.

Whole-cut success is stricter. The source cores must be recombined with the parallelity bundles and their annular gluing data. The full audit obtains 111 successes rather than 118. This is why a report of "106 out of 106" without saying *cores* would be misleading.

The nine new whole-cut packets are listed in Table 2. The notation g^m means m components of genus g, not a power in a numerical calculation.

There are $102 + 111 = 213$ independent positive whole-cut replays in the paired audit. The separate Regina literal-cut oracle completes on 113 of the 120 inputs. It agrees with all 104 deferred positives among those 113 cases. The other seven native positives exceed the oracle's 4,096-normal-disc expansion cap; their native certificates still replay, but no literal comparison is claimed. No oracle timeout or discrepancy was observed in this run.

Of the nine remaining deferred refusals, eight contain a component that the literal oracle identifies as a nonhandlebody. The remaining example, `pilot/finite_figureEight/0`, is a

Source identifier	Census	New kernel v3 used?
`pilot/fibonacci_lst_04/0`	$1^1, 2^1$	No
`pilot/fibonacci_lst_06/0`	$1^1, 2^1$	No
`pilot/fibonacci_lst_08/0`	$1^1, 2^1$	Yes
`pilot/fibonacci_lst_10/0`	$1^1, 2^1$	Yes
`pilot/lst_4_9/0`	$1^1, 2^1$	No
`pilot/finite_trefoil/0`	2^1	No
`pachner23_5_8_2/mixed`	$0^1, 1^2$	No
`interior_loop/empty`	1^1	Yes
`interior_loop/boundary_link`	$0^1, 1^1$	Yes

Table 2: Nine new complete whole-cut source packets. Other positive paths use existing core/absorption schemas with supplied forests.

union of genus-one and genus-two handlebodies. It is an explicit incompleteness witness for the present whole-cut certificate strategy. The eight other refusals are consistent with the oracle but are still only `INCONCLUSIVE` in the native API.

8.2 Finite exhaustive audits

The exceptional selector was compared against an independent oracle that enumerates every spanning tree and peels using literal, uncapped integer column totals. The oracle imports neither the selector's tree routines nor its peeling backend. It independently replays every positive output.

Finite instance class	Instances
Exhaustive three-row, two-column cases	46,656
Seeded general weighted cases	2,000
Seeded cycle-compatible column cases	2,000
Additional edge cases	2
Total	50,658
Successful trees, independently replayed	5,183
No allowed successful tree	45,475
Mismatches	0

Table 3: Standalone finite selector audit. Random matrices are not asserted to have native geometric realizations.

The exhaustive family has three row edge records chosen from a null record, a root loop, a nonloop joining two vertices, and a loop at the other vertex; six matrix entries range over $\{0, 1, 2\}$. It therefore has $4^3 3^6 = 46{,}656$ instances. Disconnected graphs test the input rejection path as well as connected instances testing the theorem. The audit also checks partial-partition counts for $u = 0, \ldots, 6$:

$$1,\ 2,\ 5,\ 15,\ 52,\ 203,\ 877,$$

and two zero-budget cases. Its recorded runtime is an audit detail, not evidence for a worst-case exponent.

The deferred primal tests independently enumerate every spanning forest on 120 seeded small closed-walk instances and compare existence of a successful forest with full-matrix peeling. Native tests cover the strict source gains, loops, repeated bridge traversals, the

closed-walk hypothesis, and rejected certificate mutations. The geometric family proof is separate from these finite checks.

8.3 Regression accounting

All 1,707 distinct real tests in the restored baseline-plus-patch test inventory pass across the main run and targeted recovery. This is not reported as one clean invocation: the initial sparse checkout omitted historical fixtures and reference modules. Its 1,620 reported tests included eleven failed-import placeholders; there were also 29 fixture-dependent method errors and three failed class setups.

Restoring the required tracked directories resolved those dependencies. The 24 affected modules contain 174 tests. Three CLI tests initially failed in the recovery harness because its working directory did not propagate to their subprocess import paths; those three passed when run from the documented `fast` directory. No production edit was needed. There were 1,580 initial real passes, 174 recovered tests, and 47 overlaps, giving $1{,}580 + 174 - 47 = 1{,}707$ distinct passes. A discovery-only inventory confirms that count and zero loader errors. All eleven new test methods are included. Logs and exact commands retain the unsuccessful environmental runs as well as the successful recovery.

8.4 Performance: stronger search does have a local cost

Final timings were run after test activity finished, with CPython 3.12.14, Regina 7.4, and Linux 6.18.44 on x86-64. These are measurements in one runtime environment, not statistically representative performance claims. Raw samples and all timer boundaries are supplied.

For the 110 native cores on which both strategies succeed, we precompute source geometry and the dual reduction, then time nine alternating-order repeats of 25 corpus passes: 2,750 core-producer calls per measurement. The median batch time is 116.115 ms for canonical and 161.272 ms for deferred, a ratio of 1.389. Thus the stronger core producer is about 39% slower on this common-success subset. These timings exclude source preparation and certificate replay. They explain the opt-in default; they do not negate the exact elimination of primal enumeration.

For six common-success whole-cut queries, Table 4 times the complete native producer plus independent replay. Seven randomized-order pairs are used per case, followed by three deferred/deferred A/A pairs. Short timings show visible variability; both medians and median paired ratios are retained, which need not equal the ratio of the two separate medians.

Common-success source	Canonical ms	Deferred ms	Paired D/C
`fibonacci_lst_01`	2.066	2.317	1.104
`fibonacci_lst_03`	5.073	5.075	0.989
`fibonacci_lst_05`	58.498	65.774	1.017
`fibonacci_lst_07`	67.588	70.509	1.045
`lst_0_1`	14.816	16.235	1.003
`lst_1_5`	28.618	30.770	1.005

Table 4: Complete native-query medians. Identifiers abbreviate `pilot/<name>/0`. A ratio above one favors canonical.

For new successes, a canonical refusal is not a comparable successful operation. Instead we compare deferred production plus replay with Regina source construction, explicit cutting, and recognition of every component, starting from the same raw input. Table 5 shows that the

native path is slower on five of these six small expanded inputs. The trefoil-related cut favors the native path. These results support no blanket claim of a faster wall-clock recognizer; the benefits are a larger certified compressed class and a proved removal of a search variable.

New-success source	Deferred ms	Explicit ms	Paired D/E
`pilot/fibonacci_lst_04/0`	42.258	1.963	25.880
`pilot/fibonacci_lst_08/0`	69.602	6.593	10.720
`pilot/fibonacci_lst_10/0`	86.238	9.897	8.650
`pilot/finite_trefoil/0`	5.632	28.922	0.190
`interior_loop/empty`	3.327	0.634	5.421
`interior_loop/boundary_link`	6.791	1.081	6.278

Table 5: Same successful whole-cut census query, different operations. Explicit means Regina's expanded cut and component recognizer. No complete knot-diagram recognition is timed.

8.5 The all-size forest family

At $h = 10$, the specified unpruned lexicographic baseline examines exactly 29,525 forests among 59,049 candidates before its first success. It took 882.154 ms; deferred core discovery averaged 0.05655 ms over 101 calls. The theorem, not the timing ratio, establishes the asymptotic separation from that particular enumeration. A block-aware algorithm also solves this family efficiently.

The saved driver separately scales deferred discovery through six sizes,

$$h = 16, 64, 256, 1024, 4096, 16384,$$

storing five timings at each size. Its queue visits exactly $5h$ incidences, performs no fill updates, and emits $2h$ pivots. Allocation and cache effects make wall-clock measurements nonuniform; an indexed linear-work proof is stronger evidence than fitting a slope to those six points. The family has a genuine handle realization, but the benchmark consumes explicit graph walks and does not authenticate a native triangulation.

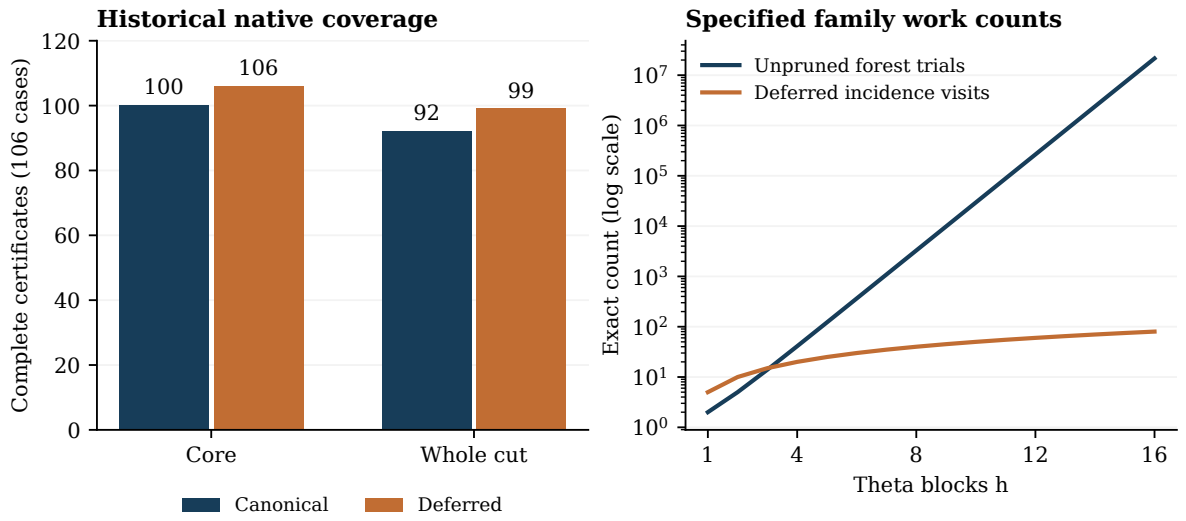


Figure 1: Left: exact historical certificate counts, separating cores from whole cuts. Right: the proved number of forests tested by the specified unpruned family enumerator and the number of deferred incidence visits. The right panel compares exact work counts of different kinds; it is not a universal lower bound or a wall-clock speedup chart.

9 A terminal recognition route through free fundamental groups

The raw cancellation algorithms deliberately recognise a restricted certificate class. A different terminal certificate can use the fundamental group, provided that irreducibility is already justified by the source topology. This permits algebraic certificates to establish an *abstract* handlebody conclusion without interpreting the algebraic moves as geometric handle slides. We work with compact PL manifolds and locally flat, properly embedded surfaces; equivalent smooth conventions give the same statements.

Lemma 9.1 (Cutting along surfaces with boundary)**.** *Let M be a connected irreducible compact orientable three-manifold. Let S be a finite disjoint union of compact, properly embedded, two-sided surfaces such that every connected component of S has nonempty boundary. Then every component of $M \setminus\setminus S$ is irreducible. No incompressibility hypothesis on S is required.*

Proof. An embedded sphere Σ in the interior of a cut component gives an embedded sphere in $\operatorname{int} M$ disjoint from S. It bounds an embedded ball B in M. Since ∂B misses S, every connected component of S that meets B must lie entirely inside B. Such a component cannot meet ∂M, contradicting its nonempty boundary and proper embedding. Thus $B \cap S = \varnothing$. Choosing the regular neighbourhood used for cutting sufficiently small, B lies in the cut component containing Σ. Changing this neighbourhood only changes the cut by a collar isotopy, so the conclusion holds for the given cut. □

The hypothesis about components is necessary for this argument. Cutting a ball along an interior sphere produces a component homeomorphic to $S^2 \times I$, which is not irreducible. If closed cutting surfaces are allowed, irreducibility of their complementary pieces needs a separate justification.

The next implication is classical; see Hempel [3]. We include its standard compression argument to make the precise hypotheses and the terminal use explicit.

Proposition 9.2 (Classical free-group handlebody criterion)**.** *Let N be a compact connected orientable irreducible three-manifold with $\partial N \neq \varnothing$. If $\pi_1(N) \cong F_g$, the free group of rank $g \geq 0$, then N is a handlebody of genus g. In particular, $g = 0$ implies that N is a three-ball.*

Proof. If a boundary component F has positive genus, the homomorphism $\pi_1(F) \to \pi_1(N)$ cannot be injective: subgroups of free groups are free, whereas a closed orientable positive-genus surface group is not free. The Loop Theorem therefore supplies a properly embedded disk $D \subset N$ with ∂D essential in F.

Cutting along D preserves irreducibility by Lemma 9.1. Van Kampen's theorem gives $\pi_1(N) = \pi_1(N_1) * \pi_1(N_2)$ when D separates, and $\pi_1(N) = \pi_1(N \setminus\setminus D) * \mathbb{Z}$ when it does not. Consequently all cut components again have free fundamental group. Every component still has nonempty boundary.

This process terminates. Indeed, for the current union W of components define

$$\mu(W) = \sum_{\substack{F \subset \partial W \\ g(F) > 0}} (2g(F) - 1).$$

Compression along an essential separating curve on a boundary component decreases μ by one. Compression along a nonseparating curve in that component decreases it by two, except on a torus, where it decreases it by one. Thus every disk cut strictly decreases this nonnegative integer.

At termination all boundary components are spheres. A connected irreducible three-manifold with a sphere boundary component is a ball: push that sphere slightly into the interior and use the ball it bounds on the side away from its boundary collar. Hence all terminal components are balls. Reversing the disk cuts attaches one-handles to these balls, and recovers a handlebody. Its genus is g, since a genus-g' handlebody has fundamental group $F_{g'}$ and free-group rank is an isomorphism invariant. □

Corollary 9.3 (Source-bound terminal group certificates)**.** *Under the hypotheses of Lemma 9.1, assume additionally that $\partial M \neq \varnothing$, and let N_i be the connected components of the* whole *cut. Suppose a source verifier reconstructs a presentation $\mathcal{P}_i$ of $\pi_1(N_i)$, and an algebraic verifier certifies*

$$\mathcal{P}_i \;\cong\; \langle y_1, \ldots, y_{g_i} \mid \rangle$$

for every i. Then the cut is a disjoint union of handlebodies of genera g_i. It is a disjoint union of balls exactly when every g_i is zero.

Proof. Every N_i is irreducible by the lemma, and has nonempty boundary because the source and cutting surfaces have boundary. The verified group isomorphism and Proposition 9.2 give the conclusion component by component. □

One possible algebraic certificate is a sequence of elementary Tietze transformations from the authenticated presentation to the displayed free presentation. The checker must verify each transformation. A relator claimed to follow from earlier relators must carry an explicit consequence witness, such as a product of conjugates; its validity cannot be delegated to an unspecified group-word oracle. Generator definitions, relator inversions, conjugations, products, and their certified inverses give familiar checkable building blocks. If words are compressed, their equality checks and all certificate sizes must be measured in that encoding. The initial presentation must describe the entire component, including bundle gluing relations, rather than merely its source core.

This is a proposed terminal extension, not an additional implemented success in the reported experiments. Its use requires authenticated irreducibility and the stated source identification. It supplies neither a meridian disk system in marked coordinates nor a boundary-pattern transport map. A positive free rank gives a handlebody rather than the ball required at the end of an essential hierarchy; a boundary-pattern essentiality test remains separate even when the underlying piece is a ball. Finally, the classical theorem and the existence of a finite Tietze certificate provide no polynomial or quasi-polynomial bound on finding such a certificate, or on its size. These are separate research questions.

10 What would actually imply quasi-polynomial recognition?

The new results remove a local combinatorial obstruction to efficient normalization. They do not bound the number, size, or required geometry of the states visited by a complete knot algorithm. This section makes the distinction quantitative, so that future work has a falsifiable complexity target rather than an analogy with a fast local benchmark.

10.1 A conditional composition theorem

Proposition 10.1 (Explicit accounting for a complete implementation)**.** *Suppose a correct total unknot-recognition implementation on diagrams of size n makes at most $Q(n)$ calls to its geometric state-processing operator. Suppose all state encodings have size at most $N_*(n)$, every exceptional interface has size at most $U(n)$, and all remaining work per state, including certificate generation, geometric transport, and replay, is bounded by a fixed polynomial in $N_*(n)+1$. Allow at most $(N_*(n)+1)^{O(1)}$ selector invocations per state, each on an input of encoded size $(N_*(n)+1)^{O(1)}$. Assume also that all work outside these state calls, including preprocessing and finalization, is polynomial in $n+N_*(n)+1$. If it uses the exact finite selector of Theorem 5.5, then its total time is bounded by*

$$T(n) \le (Q(n)+1)\,(n+N_*(n)+1)^{O(1)} 2^{O(U(n)\log(U(n)+2))}. \tag{11}$$

In particular, the three bounds

$$\begin{aligned} \log(Q(n)+2) &= O((\log(n+2))^2), \\ \log(N_*(n)+2) &= O((\log(n+2))^2), \\ U(n)\log(U(n)+2) &= O((\log(n+2))^2) \end{aligned}$$

suffice for $T(n) = n^{O(\log n)}$ for $n \ge 2$.

Proof. Each finite selection costs at most a polynomial in its encoding times the Bell-number factor. The input-size hypothesis and the bound on the number of invocations absorb all selector calls into the same fixed polynomial factor per state. Sum this and the remaining per-state costs over at most $Q(n)$ calls, then absorb the outside work using $Q(n)+1$ and the polynomial in $n+N_*(n)+1$. Taking logarithms gives terms bounded by $\log(Q(n)+2)$, $O(\log(n+N_*(n)+1))$, and $O(U(n)\log(U(n)+2))$. Each is $O((\log n)^2)$ under the stated hypotheses. Exponentiating gives the conclusion. □

This is complexity accounting, not a new total-recognition theorem. The first sentence assumes correctness and termination of a complete implementation. Our existing partial census procedure does not satisfy that premise merely because every emitted certificate is correct.

A more convenient sufficient state-size condition is $N_*(n) = n^{O(1)}$. Under this stronger condition, $U = O(\log n)$ gives local cost

$$2^{O(\log n \log\log n)} = n^{O(\log\log n)}.$$

The weaker interface bound $U = O((\log n)^2/\log\log n)$ also suffices for the target local $n^{O(\log n)}$ estimate, away from small n. Neither interface bound is proved for arbitrary native inputs or along the required hierarchies. The absence of a dependence exponential in the number of top handles does not itself control U.

10.2 Hierarchy depth, loop counts, and the displayed exponent

The loop estimate $L(g+1)^L$ from Proposition 9.1 of [6] uses the maximum hierarchy length L and the maximum *pattern complexity* g of its decomposing surfaces. This complexity includes boundary-pattern intersections; a genus bound alone does not control it. The estimate immediately shows why quantitative depth matters. If $g \le n^a$ and $L \le b\log n$, then

$$\log\bigl(L(g+1)^L\bigr) = \log L + L\log(g+1) = O((\log n)^2).$$

If instead $L = O((\log n)^2)$ with the same pattern-complexity bound, this calculation gives $O((\log n)^3)$ in the exponential, namely $n^{O((\log n)^2)}$. A bound merely linear in a handle-complexity parameter that can itself be polynomial in n does not give either logarithmic-depth conclusion.

The purpose of comparing the two 2021 slide estimates is therefore specific: an implementation and proof should state which exponent they establish. Replacing a polynomial local factor by a faster polynomial does not repair an uncontrolled hierarchy depth. Conversely, a quasi-polynomial number of states can still tolerate another quasi-polynomial local factor of the same logarithmic exponent, as Proposition 10.1 shows.

10.3 Encoding is part of the theorem

Let t count the source tetrahedra, and let b be the total bit length of the supplied normal coordinates. A local source-to-certificate bound should be expressed in $t + b$ and in the encoded marked interfaces it creates. It should not hide a scan through W parallel sheets behind a phrase such as "construct the cut." The distinction is familiar from the orbit-counting machinery of Agol–Hass–Thurston [1]; its algorithmic value is that large multiplicities can be processed through finite interval data.

The deferred theorem never requires expanding those sheets: it operates on a supplied finite incidence matrix. But this does not prove that every matrix needed later in a hierarchy will have small encoded size. Annular paths, port refinements, boundary patterns, and repeated chart substitutions can dominate the size even when scalar multiplicities remain in binary. Every successor operator therefore needs an explicit bound on both the number of records and the bits in each record.

Output size creates an independent barrier. If a cut has exponentially many identical ball components, writing one certificate per component cannot take polynomial time in a binary multiplicity. The existing census representation groups such components by genus and multiplicity. A full hierarchy certificate needs an equally explicit policy for sharing repeated local structures and for authenticating that sharing.

10.4 Outstanding geometric and algorithmic obligations

The known local hierarchy theorems cited in Section 1 are valuable starting points. The following are the stronger or implementation-specific obligations still absent from this package.

1. *Controlled progress.* Produce the decomposing multisurfaces or other accelerated progress objects needed for a short global construction, and prove their depth and pattern-complexity bounds, including boundary-pattern intersections, in the actual source encoding. Finding an arbitrary nonseparating surface is not the open subproblem here.
2. *Source-authenticated successor transport.* Implement the applicable compressed cutting and parallelity constructions with explicit marked boundary maps, preserving the hypotheses needed for the next state. A census is less information than a successor handle structure.
3. *Repair and restart accounting.* Bound all branches caused by unsuccessful local candidates, changes of charts, boundary-pattern repairs, or restarts. Charging only successful cuts omits real work.
4. *A complete terminal procedure.* The raw certificate class is not complete for all handlebodies in arbitrary charts. A total algorithm must explain what happens after a universal raw obstruction, and bound the cost of its alternative. It must also decide the relevant boundary-pattern condition, not just whether the underlying pieces are balls.
5. *Exceptional-interface control.* Bound u, replace it by a more stable parameter, or find a polynomial selector for the actual geometric class. The abstract triangle witness prevents one particularly simple invariance argument but does not settle that geometric question.

These obligations interact. A move that reduces u while making marked paths exponentially long may lose overall. A normalization that reduces the number of records but changes the admissible pattern may invalidate the next theorem. The intended measure must charge both geometry and encoding, with a source verifier checking each claimed transition.

10.5 Algebraic simplification and geometric transport

Whitehead minimization has polynomial algorithms in the length of explicitly written free-group words [15]. Such a bound is not automatically polynomial in the size of a compressed grammar for those words. More fundamentally, an algebraic automorphism does not by itself give the marked meridian and port data expected by the native geometric kernel. The recent thickenable-presentation work [7] provides relevant structural context, but no bounded marked-slide implementation is supplied here.

Section 9 gives a legitimate alternative use: under authenticated irreducibility hypotheses, a verified free fundamental group proves an abstract terminal handlebody. It can bypass the need to interpret each algebraic transformation as a slide when only that terminal conclusion is required. It still needs a bound on the source presentation, the certificate, and its discovery. It also cannot be substituted for marked transport when the algorithm must continue.

11 A proposed research programme

The most useful next steps should either remove a named asymptotic obstruction or produce an independently checkable new geometric success. The questions below are ordered by their likely value to this codebase. They are proposals, not conjectures silently used in the proofs above.

11.1 Immediate implementation and diagnostic work

1. Authenticate the exceptional selector on native inputs. Build the missing adapter for Corollary 6.1. Export the full pre-primal occurrence matrix and source dual endpoints, mark nonselectable rows, run the finite selector, and feed the resulting tree through the existing geometric reduction. Reuse the ordinary source verifier for every positive packet. The initial acceptance criterion is correct transport and equality with exhaustive dual-tree enumeration on small native cores, followed by a source-authenticated example where dual choice adds coverage after deferred primal selection. A synthetic matrix success is insufficient for this milestone.

2. Resolve the remaining literal handlebody example. The fixture `pilot/finite_figureEight/0` is the only all-handlebody whole cut missed among the 113 completed literal oracle comparisons. Its components have genera one and two. Determine the exact obstructing annular or augmented relation and produce the smallest checkable repair. First establish which geometric and irreducibility hypotheses apply; then compare a supported slide certificate, a group-based terminal certificate, or a change of dual reduction where one exists. This is a concrete native target for moving beyond raw stopping sets.

3. Implement an adaptive canonical/deferred policy. Can the canonical fast path be retained without paying twice for source construction when it stalls? Share immutable full rows and source component data; run canonical peeling as a view or maintain enough deleted-column metadata to retry deferred discovery. Measure source preparation, producer time, replay, allocation, and certificate size separately. The target is to preserve the historical common-success latency while retaining all nine new whole-cut results. Keep the simple opt-in path available as a reference implementation.

4. Export and minimize universal primal obstructions. The greatest stopping residual is canonical for a fixed matrix, but it need not be the smallest witness. Find small source-bound stopping subsets for diagnosis and caching. Determine whether minimum-cardinality or minimum-support stopping witnesses are tractable in the actual native class. A heuristic minimizer is useful if an independent checker verifies its result. Every cache key must retain source identity, retained rows, and the raw occurrence convention.

11.2 Sharper finite algorithms

5. Replace Bell enumeration by representative families. The completeness proof needs at most u additional tree witnesses. Can a representative-family construction in the graphic matroid preserve all useful interface signatures and reduce the $2^{O(u \log u)}$ factor to $2^{O(u)}$, or give a small kernel? Any such reduction must preserve both nonlinear Horn closure and simultaneous independence with the mandatory representatives. Keeping one edge per signature is not obviously safe: equal activation signatures can have different graphic dependence with the chosen forest.

6. Parameterize the compressed rule interaction instead of u. Construct the bipartite graph of exceptional rules and interface rows, or the incidence graph after baseline strongly connected components are contracted. Investigate dynamic programming by treewidth, branchwidth, or separator size while retaining the graphic-independence constraint. A useful theorem should specify its states and account for connectivity partitions; otherwise a hidden

Bell-number factor can reappear in each bag. The native cycle condition may constrain this graph more strongly than arbitrary hypergraph activation.

7. Settle the genuinely native triple-column problem. The saved nine-vertex witness refutes tree invariance from sparsity and dual-cycle parity alone. Which additional constraints imposed by an actual triangulated normal cut exclude it, if any? Either construct a native realization preserving the required ports, or prove a stronger geometric restriction and exploit it algorithmically. This would decide whether the remaining selector should be studied as a general constrained activation problem or as a substantially narrower topological class.

8. Make exhaustive dual failure independently checkable. For one partial partition, matroid intersection can expose a rank defect. Can the failures of all partitions be compressed into a small certificate that covers the entire search space, perhaps by shared Horn-closure states and reused rank inequalities? The objective is a certificate for failure of the finite raw model, not a nonunknot certificate. A worst-case size bound would be more valuable than merely logging the full exponential enumeration.

9. Analyze the best parameter after legal simplification. The theorem uses exceptional rows in the input before unknown dual deletions. Study the minimum achievable exceptional size under cheaply certified preprocessing, preserving the exact forest-selection problem. Possible reductions include automatic-seed closure, domination of Horn rules, irrelevant-column removal, and decompositions with independently tracked graphic constraints. Prove equivalence for each reduction; deleting a row because it would become useful in some unknown tree is not valid preprocessing.

11.3 Geometry beyond the current raw model

10. Certify marked slides that decrease an explicit potential. Define a potential combining raw intersection size, exceptional support, and encoded port length. Seek a class of geometric one/two-handle slides for which a verifier reconstructs the actual band and proves a controlled decrease or bounded amortized increase. The theta family shows that forest choice alone can suffice in some cases; the remaining whole-cut example asks for a move beyond that theorem. Algebraic relator shortening is a guide, not the marked geometric certificate.

11. Develop the terminal free-group certificate route. For cuts satisfying the boundary and irreducibility conditions of Section 9, construct authenticated whole-cut presentations and short Tietze or Nielsen certificates of freeness. Determine which existing compressed words admit polynomial equality checking and whether the certificate construction stays within that encoding. Separate three questions: proving the group is free, recovering an abstract genus, and recovering a marked disk system. Only the first two are needed for the terminal implication proved here.

12. Realize the deferred-basis theorem geometrically. The represented-matroid extension preserves column span but does not make every basis a collapsible subcomplex. Identify a class of regular cell structures in which the discovered basis has a certifiable geometric realization, or characterize the additional obstruction. This may connect the graph proof to higher boundary operators without confusing homological independence with simple-homotopy or handle cancellation data.

11.4 The global quasi-polynomial programme

13. Prove an interface budget along accelerated hierarchies. Can each maintained state be transformed, with bounded marked cost, to one satisfying $u \log(u+2) = O((\log n)^2)$? A weaker amortized theorem bounding the sum of per-state costs could also suffice. Measure u together with handles, ports, coordinate bits, genus, and source-grammar size on successive authentic cuts before conjecturing a logarithmic law. A logarithmic observation in a small corpus is not an asymptotic bound.

14. Build the complete marked successor operator. Implement the applicable compressed successor constructions as a total source-to-source certificate step under their geometric hypotheses. Record handles, boundary patterns, attaching maps, inherited markings, parallelity data, and all reindexing maps. Prove a uniform encoding bound under composition. This is the central bridge from successful terminal censuses to an actual hierarchy engine.

15. Bound depth, repairs, and terminal completion together. Formulate one theorem with all quantities in Proposition 10.1: the state-call count $Q(n)$, encoding size $N_*(n)$, exceptional parameter $U(n)$, and a complete terminal procedure. Surface construction should target the stronger controlled progress needed for the announced speed-up. Include unsuccessful candidate work and boundary-pattern repairs in the accounting. The decisive milestone is a proved bound for a correct total algorithm, not a collection of separately fast routines.

16. Formalize the finite invariants before the full topology. The stopping-set invariant, deferred graph forest theorem, represented binary-matroid extension, and compressed Horn reduction are finite, explicit statements with short constructive proofs. They are good initial targets for proof-assistant formalization or small verified checkers. Formalizing these layers would reduce the trusted combinatorial base while geometric source-authentication proofs continue separately. The supplied counterexamples and mutation cases are regression fixtures for that formalization, not substitutes for it.

The recommended order is to complete the native adapter and adaptive implementation while pursuing the exceptional-interface and marked successor questions in parallel. The article's strongest established asymptotic gain is already unconditional within its finite geometric model: primal forest enumeration is unnecessary. The next global advance must control the state evolution that feeds those local models.

A Package contents and reproduction

The package is an integration bundle for the source snapshot [13]. It contains the article's complete TEX source, compiled PDF, bibliography, and figure source; an additive patch and six-file overlay; the independent finite research operator; source-bound positive packets; raw audit/timing data; and reproduction commands. The repository itself and third-party papers are not duplicated.

A.1 Build the article

From the package root:

Directory or file	Purpose
`article/`	Master TEX, section files, bibliography, PDF, and figure.
`patches/`	Patch against the pinned baseline.
`overlay/`	The same six changed source/test files, with repository-relative paths.
`baseline/`	Original versions of the four changed production modules.
`finite_selector/`	New exceptional solver, attributed prior backend, independent exhaustive checker, and triangle witness.
`certificates/`	Nine whole-cut and eight core packets, retaining their source inputs.
`tools/`	Audit, replay, benchmark, and plotting drivers.
`results/`	Raw measurements, complete audit counts, regression recovery, and source validation records.
`SOURCE_AUDIT.json`	Baseline, hashes, environment, and change provenance.
`CLAIMS.json`	Machine-readable result and limitation ledger.
`SHA256SUMS`	Integrity hashes for the delivered files.

Table 6: Integration bundle layout.

```
cd article
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  Deferred_Forests_and_Exceptional_Incidence.tex
```

The supplied `build.sh` performs this command. The figure PDF is already included. Regenerating it with `tools/generate_figures.py` requires Matplotlib; compiling the article does not require Python.

A.2 Apply and exercise the native change

Check out the pinned revision or inspect the patch carefully against a later tree. From the repository root, using an absolute package path:

```
git apply --check /path/to/package/patches/deferred_forests.patch
git apply /path/to/package/patches/deferred_forests.patch
cd Topology/UnknotRecognition/fast
python3 -m unittest tests.test_deferred_forest \
  tests.test_deferred_boundary_kernel -v
```

These tests require the source fixtures and Regina used by the native project. Full discovery also needs the historical reference modules documented in the recovery log; a checkout containing only `fast` is not sufficient for that broader suite.

An actual call uses the same source data as before:

```
result = normal_cut_handlebodies(
    triangulation, coordinates,
    record_certificate=True,
    forest_strategy='deferred',
)
if result['status'] == 'COMPLETE':
```

```
    assert verify_normal_cut_handlebodies(
        triangulation, coordinates, result['certificate'])
```

The import statements and a runnable packet-replay command appear in the package README. Callers must retain the `INCONCLUSIVE` distinction.

A.3 Reproduce the finite theorem audit

No topology library is required for the standalone operator:

```
cd finite_selector
python3 audit_exceptional.py --output audit_results_new.json
python3 check_triangle_witness.py
python3 exceptional_forest.py examples/triangle.json
```

The complete audit seeds, instance classes, literal oracle, and budget checks are recorded in its source and README. Native whole-cut replay and timings have explicit `-fast-root` arguments so they can be run against an isolated patched checkout. The article's values are fixed measurements; a new run should write a new result file instead of silently replacing the delivered evidence.

A.4 Reproducibility limits

The test suite is extensive but not a formal proof of implementation correctness. The mathematical proof is independent of the observed sample counts. The standalone audits range over finite graph/matrix instances and do not establish native geometric realizability. The literal oracle comparisons exclude inputs beyond their expansion cap. The source-bound packets and raw data allow each of these claims to be rechecked separately, with their original scope intact.

B Complete data for the triangle sensitivity witness

This appendix specifies the finite object in Proposition 5.9 without requiring an external data file. The graph vertices are $0, \ldots, 8$. Rows are the 25 edge labels below. Each of the 17 Boolean matrix columns has value one on its listed three rows and zero elsewhere.

Row	Endpoints	Row	Endpoints	Row	Endpoints
0	$(0,1)$	9	$(1,8)$	18	$(4,7)$
1	$(0,2)$	10	$(2,4)$	19	$(4,8)$
2	$(0,3)$	11	$(2,5)$	20	$(5,6)$
3	$(0,4)$	12	$(2,6)$	21	$(5,7)$
4	$(0,6)$	13	$(2,7)$	22	$(5,8)$
5	$(0,7)$	14	$(3,4)$	23	$(6,8)$
6	$(1,2)$	15	$(3,7)$	24	$(7,8)$
7	$(1,6)$	16	$(3,8)$		
8	$(1,7)$	17	$(4,6)$		

Table 7: Dual graph edge records.

Column	Supported rows	Column	Supported rows
0	$\{0,1,6\}$	9	$\{11,12,20\}$
1	$\{0,4,7\}$	10	$\{11,13,21\}$
2	$\{1,3,10\}$	11	$\{14,15,18\}$
3	$\{2,5,15\}$	12	$\{15,16,24\}$
4	$\{3,4,17\}$	13	$\{17,19,23\}$
5	$\{3,5,18\}$	14	$\{18,19,24\}$
6	$\{6,8,13\}$	15	$\{20,22,23\}$
7	$\{7,9,23\}$	16	$\{21,22,24\}$
8	$\{8,9,24\}$		

Table 8: All 17 triangle columns, in the order used by the stopping totals.

The successful and failing trees, respectively, are

$$T_+ = \{5,7,10,14,17,19,21,22\},$$
$$T_- = \{1,2,5,8,12,16,19,20\}.$$

A successful trace on the complement of T_+ consists of the following $(\text{row},\text{column})$ pairs in order:

$$(23,13),\ (9,7),\ (20,15),\ (24,16),\ (8,8),\ (18,14),$$
$$(3,5),\ (1,2),\ (4,4),\ (0,1),\ (6,0),\ (13,6),$$
$$(11,10),\ (12,9),\ (15,11),\ (2,3),\ (16,12).$$

For T_- the recorded legal prefix is

$$(15,3),\ (11,9),\ (24,12),\ (9,8),$$
$$(18,14),\ (3,5),\ (10,2),\ (14,11).$$

The remaining rows are $\{0,4,6,7,13,17,21,22,23\}$. Directly summing the displayed columns on this set gives

$$(2,3,0,0,2,0,2,2,0,0,2,0,0,2,0,2,2),$$

a nonempty stopping set with no unit total. Checking each three-edge column against the endpoint table verifies that it is a graph triangle. The independent checker in the package repeats precisely these finite checks, without invoking the exceptional solver or a topology library.

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